Gurveer Madurai

+1 (778) 363-6885 | maduraigurveer@gmail.com | github.com/gurverm | linkedin.com/in/gurveer-madurai/

Education

University of British Columbia, Master of Data Science 2026
Coursework: Machine Learning, Web and Cloud Computing, Statistics, Data Structures and Algorithms

University of Toronto, Software Development Certificate 2023
Focus on Full Stack Development

University of British Columbia, Bachelor of Science 2020
Major: Biochemistry | Coursework: Data Science & Statistics, Physics, Calculus (I, II, III)

Experience

Software Engineer
Vintality Tech Inc | Hybrid, BC Jun 2024 – Aug 2025

- Architected and deployed 10+ real-time IoT dashboards, providing stakeholders with critical visibility into a network of 100+ industrial devices.
- Engineered an anomaly detection algorithm to clean and process 200K+ time-series data points, significantly improving data integrity for predictive maintenance modeling.
- Developed a high-throughput API with 5 RESTful endpoints, successfully ingesting 50K+ data points daily across 1,400+ concurrent requests.
- Collaborated on custom UI widgets and data visualization components, streamlining the monitoring of complex sensor data for end-users.

Behavior Interventionist
Vista Academy | Kelowna, BC Nov 2020 – Aug 2023

- Executed systematic data collection on behavioral patterns, utilizing quantitative tracking to monitor progress and identify environmental triggers.
- Analyzed behavioral datasets to perform root-cause analysis (ABC data), enabling data-driven adjustments to individualized intervention plans.

Projects

Data Science - Capstone UBC LiteFarm | Vancouver, BC

- Developed a data pipeline to benchmark farm crop yields against nearby farms using geospatial proximity queries and PostgreSQL.
- Built interactive visualizations and KPI summaries to help farmers compare crop performance against anonymized peer benchmarks.

Multimodal pCR Prediction for Breast Cancer – 3rd Place, National Canadian Medical Datathon

- Built a multimodal Keras model combining clinical and WSI-derived features to predict NAC response; achieved 78% accuracy, 92% precision, 93% recall
- Developed end-to-end pipeline and deployed an app for real-time inference from clinical inputs and images

Book search engine:

- Built a multi-page web application that lets authenticated users search for books through the Google API and save results
- Implemented CRUD functionality, authentication, and API-backed data flows using MongoDB, Express, React, JavaScript, GraphQL, and JWT

Technologies and Languages

Languages: Python, JavaScript, SQL, R
Web: React, NextJS, NodeJS, ExpressJS, jQuery, Tailwind CSS, Bootstrap, HTML/CSS
DB and DevOps: MongoDB, InfluxDB, MySQL, CI/CD, Git, AWS Lambda, EC2, S3, Docker
Machine Learning: NumPy, pandas, scikit-learn, PyTorch, XGBoost, RAG, MCP